

# CineStream Content Analytics Dashboard

*Sem 2 OJT Self Project – Interactive Dashboard with Streamlit*

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| <b>Name</b>            | Ajeeba Farhath P.P                     |
| <b>Register Number</b> | 251811800225                           |
| <b>Class</b>           | BCA with AI and Data Science           |
| <b>Project Title</b>   | CineStream Content Analytics Dashboard |

**GitHub:** [https://github.com/ajeebafarhath-web/cinestream\\_dashboard](https://github.com/ajeebafarhath-web/cinestream_dashboard)

**Live Dashboard:** <https://cinestreamdashboard-ajeebafarhathpp.streamlit.app/>

## 1. Introduction

The CineStream Content Analytics Dashboard is an interactive web-based analytics application developed using Streamlit. The project analyzes the complete content catalog of a fictional OTT platform named CineStream, containing more than 500 titles across Movies, Series, Documentaries, and Stand-up Specials.

The main objective of this project is to transform a raw CSV dataset into an interactive dashboard that helps different business teams understand audience engagement, profitability, content quality, language performance, and content trends through interactive visualizations and filtering.

## 2. Project Objectives

The objectives of this project are:

- Analyze the CineStream content catalog
- Clean and preprocess the raw dataset
- Build an interactive Streamlit dashboard
- Provide meaningful business insights using visualizations
- Enable users to filter data dynamically
- Help business teams make data-driven decisions

## 3. Technologies Used

| Technology                | Purpose                  |
|---------------------------|--------------------------|
| Python                    | Programming Language     |
| Streamlit                 | Dashboard Development    |
| Pandas                    | Data Cleaning & Analysis |
| NumPy                     | Derived Calculations     |
| Matplotlib                | Statistical Charts       |
| Plotly                    | Interactive Charts       |
| GitHub                    | Version Control          |
| Streamlit Community Cloud | Dashboard Deployment     |

## 4. Dashboard Features

The CineStream dashboard provides the following features:

- Interactive sidebar filters

- KPI Cards showing total titles, views, watch hours, and average IMDb score
- Catalog Preview with sample data table
- Top Viewed Titles leaderboard
- Monthly Content Trend visualization
- Content Type Distribution charts
- Genre Analysis with bar charts
- Language Treemap for hierarchical views
- Production Cost vs Revenue scatter analysis
- ROI Analysis by genre
- IMDb Score Distribution histogram
- IMDb vs Views correlation scatter plot
- Smart Status Messages and alerts
- Download Filtered Dataset feature
- Cached Data Loading for performance
- Responsive Dashboard Layout

## 5. Business Questions & Analysis

### 5.1 Which genres and languages perform the best?

The dashboard shows that Reality is the highest-performing genre based on total views. The language performance analysis indicates that Spanish generates the highest average views per title, while Japanese records the lowest average views.

This suggests that CineStream should continue investing in Reality content and expand Spanish-language productions to maximize audience engagement.

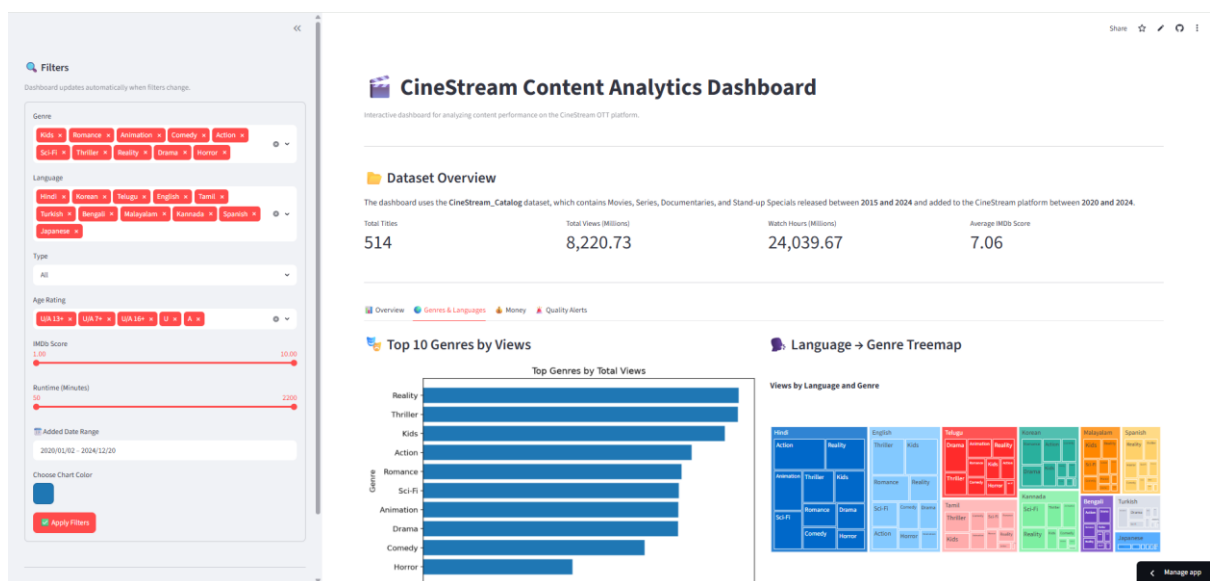


Figure 1: Genres & Languages Dashboard – Top Genres by Views and Language-Genre Treemap

## 5.2 Are high-budget productions earning profits?

The Production Cost vs Revenue scatter plot shows the relationship between production investment and revenue generated. The dashboard reports an average ROI of approximately 183.56%, indicating that the platform overall earns a healthy return on investment. However, several expensive productions still fail to recover their production costs.

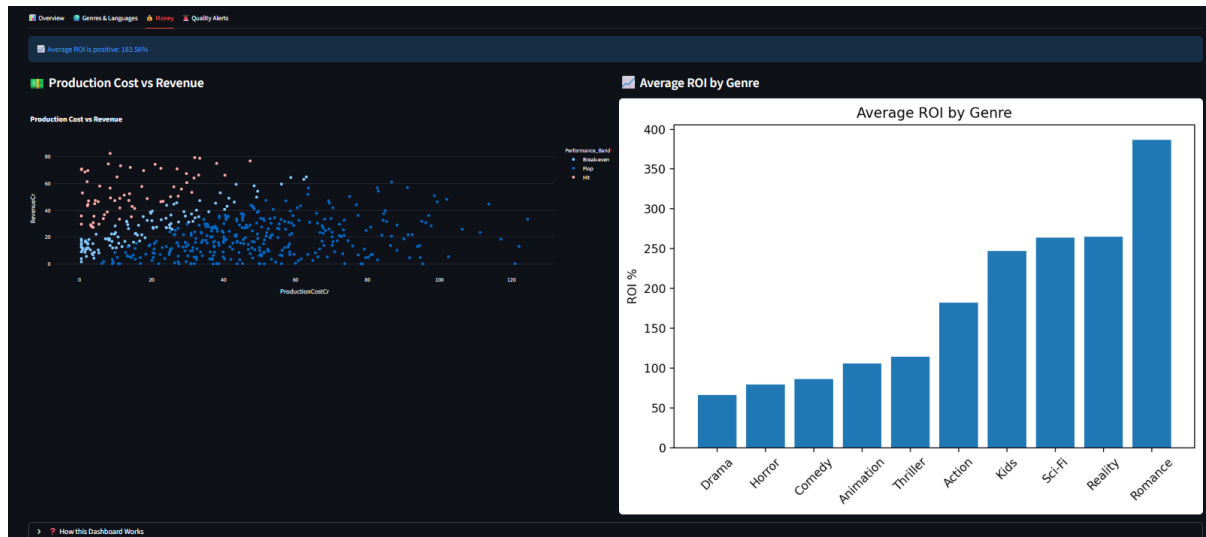


Figure 2: Money Dashboard – Production Cost vs Revenue and Average ROI by Genre

## 5.3 How is the content catalog distributed?

The dashboard provides an overview of the CineStream catalog through KPI cards, content previews, and distribution charts. The platform contains Movies, Series, Documentaries, and Stand-up Specials released between 2015 and 2024 and added to the platform between 2020 and 2024. Interactive filters allow users to explore specific content categories efficiently.

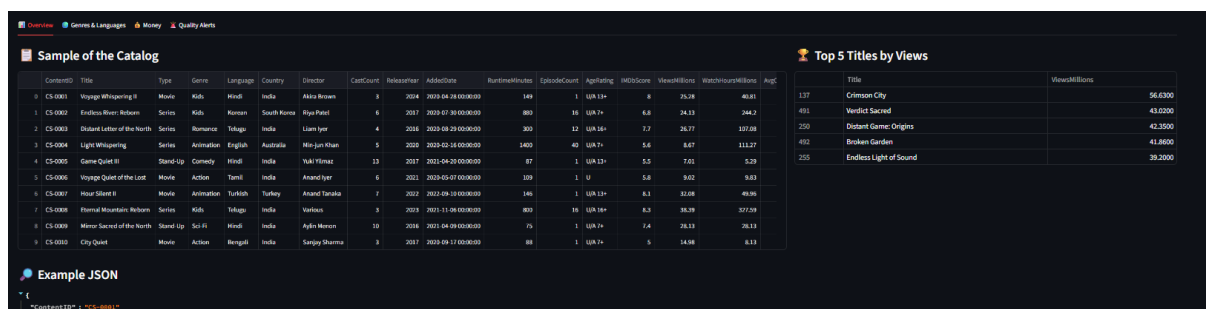


Figure 3: Overview Dashboard – Sample Catalog and Top 5 Titles by Views

## 5.4 Which titles attract the highest audience?

The dashboard identifies the most viewed titles available on the platform. The highest viewed title is Crimson City with 56.63 Million Views.

Other top-performing titles include:

- Verdict Sacred – 43.02 Million Views
- Distant Game: Origins – 42.35 Million Views
- Broken Garden – 41.86 Million Views
- Endless Light of Sound – 39.20 Million Views

These titles contribute significantly to the platform's overall audience engagement.

## 5.5 Do higher IMDb scores generate more views?

The IMDb Score vs Views scatter plot shows a generally positive relationship between audience ratings and total views. Although a few exceptions exist, titles with higher IMDb scores tend to receive greater audience attention and higher viewing numbers. This indicates that content quality positively influences viewer engagement.

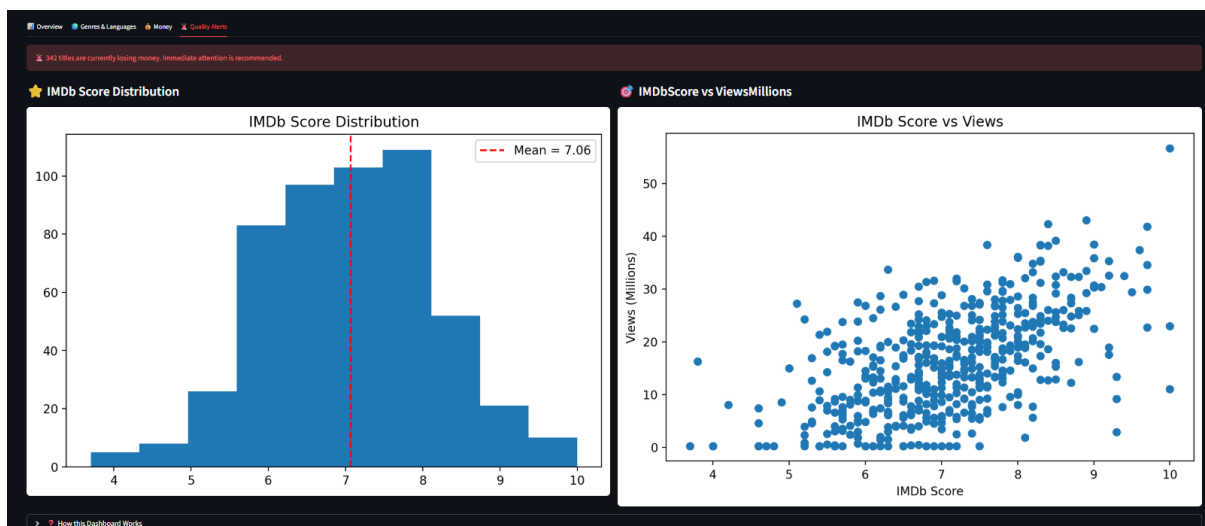


Figure 4: Quality Alerts Dashboard – IMDb Score Distribution and IMDb Score vs Views

## 5.6 Which titles are losing money?

The dashboard identifies 342 titles where Production Cost exceeds Revenue. The smart alert system automatically highlights this issue using a red status banner, helping finance teams quickly identify financially underperforming content. This insight can support better budgeting and future investment decisions.

## 6. Key Findings

- Reality is the highest-performing genre by total views
- Spanish is the best-performing language by average views per title
- Japanese has the lowest average views per title

- Average ROI across the platform is approximately 183.56%
- Crimson City is the most viewed title with 56.63 Million Views
- Higher IMDb scores generally correlate with higher audience views
- 342 titles are currently operating at a financial loss

## 7. Conclusion

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The CineStream Content Analytics Dashboard successfully converts a static CSV dataset into an interactive business intelligence solution. Using Streamlit, Pandas, NumPy, Matplotlib, and Plotly, the dashboard enables users to explore content performance through dynamic filtering and visual analytics.

The dashboard provides valuable insights into audience behavior, profitability, language performance, genre popularity, and content quality. These insights can assist marketing teams, finance departments, and management in making informed, data-driven decisions.

Overall, this project demonstrates practical skills in data cleaning, exploratory data analysis, dashboard development, data visualization, and cloud deployment using modern Python tools.

## 8. References

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- CineStream\_Catalog.csv – Primary Dataset
- Streamlit Documentation
- Pandas Documentation
- NumPy Documentation
- Matplotlib Documentation
- Plotly Documentation
- GitHub Documentation